1. Aim of the practical: Creating and managing users and groups using user add, userdel, groupadd and passwd. Controlling Access to Files with Access Control Lists (ACL) using chmod, chgrp and chown.

2. Objective: To ensure proper user and group management in a multi-user system by using appropriate commands.
To manage file permissions and control access to files for specific users and groups to ensure data security.

3. Hardware and Software Requirements:

Computer System: Host system running virtualization software.
Operating System: OS on the virtual machine (e.g., Fedora).

4. Commands/Steps:

Step-1: Creating a new user using sudo useradd username.

```
liveuser@localhost-live:~$ sudo useradd Hero
```

Step-2: Checking if the user is created using cat /etc/ passwd

```
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/usr/sbin/nologin
dbus:x:81:81:System Message Bus:/:/usr/sbin/nologin
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/usr/sbin/nologin
systemd-coredump:x:999:999:systemd Core Dumper:/:/usr/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-oom:x:998:998:systemd Userspace OOM Killer:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
qemu:x:107:107:qemu user:/:/sbin/nologin
polkitd:x:114:114:User for polkitd:/:/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/:/sbin/nologin
unbound:x:996:996:Unbound DNS resolver:/var/lib/unbound:/sbin/nologin
geoclue:x:995:994:User for geoclue:/var/lib/geoclue:/sbin/nologin
nm-openconnect:x:994:992:NetworkManager user for OpenConnect:/:/sbin/nologin
gluster:x:993:990:GlusterFS daemons:/run/gluster:/sbin/nologin
saslauth:x:992:76:Saslauthd user:/run/saslauthd:/sbin/nologin
rtkit:x:172:172:RealtimeKit:/:/sbin/nologin
pipewire:x:991:989:PipeWire System Daemon:/run/pipewire:/usr/sbin/nologin
chrony:x:990:988:chrony system user:/var/lib/chrony:/sbin/nologin
dnsmasq:x:989:987:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/usr/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
openvpn:x:988:986:OpenVPN:/etc/openvpn:/sbin/nologin
nm-openvpn:x:987:985:Default user for running openvpn spawned by NetworkManager:/:/sbin/nologin
colord:x:986:984:User for colord:/var/lib/colord:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
abrt:x:173:173::/etc/abrt:/sbin/nologin
wsdd:x:985:982:Web Services Dynamic Discovery host daemon:/:/sbin/nologin
flatpak:x:984:981:Flatpak system helper:/:/usr/sbin/nologin
gdm:x:42:42:GNOME Display Manager:/var/lib/gdm:/usr/sbin/nologin
gnome-initial-setup:x:983:980::/run/gnome-initial-setup/:/sbin/nologin
vboxadd:x:982:1::/var/run/vboxadd:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/usr/sbin/nologin
tcpdump:x:72:72:tcpdump:/:/usr/sbin/nologin
passim:x:975:975:Local Caching Server:/usr/share/empty:/usr/sbin/nologin
liveuser:x:1000:1000:Live System User:/home/liveuser:/bin/bash
john:x:1001:1001::/home/john:/bin/bash
bob:x:1002:1003::/home/bob:/bin/bash
Camper:x:1004:1006::/home/Camper:/bin/bash
developer1:x:1005:1007::/home/developer1:/bin/bash
Hero:x:1006:1008::/home/Hero:/bin/bash
```

Step-3: Now Creating a new group using sudo groupadd starsGroup

```
liveuser@localhost-live:~$ sudo groupadd starsGroup
```

Step-4: Now adding Hero user into starsGroup

```
liveuser@localhost-live:~$ sudo usermod -aG starsGroup Hero
```

```
liveuser@localhost-live:~$ getent group starsGroup
starsGroup:x:1009:Hero
```

Step-5: Now, Checking the Hero groups.

```
liveuser@localhost-live:~$ groups Hero
Hero : Hero starsGroup
```

Step-6: Now deleting user Hero and checking if its deleted.

```
liveuser@localhost-live:~$ sudo userdel -r Hero
```

```
liveuser@localhost-live:~$ groups Hero
groups: ‘Hero’: no such user
```

Step-7: Now deleting the starsGroup and checking if its deleted

```
liveuser@localhost-live:~$ sudo groupdel starsGroup
groupdel: group 'starsGroup' does not exist
```

Step-8: touch asr.text command is used to create an empty file named asr.text.

```
asr@fedora:~$ touch asr.text
```

```
asr@fedora:~$ ls
asr.text  Documents  Music     Public     Videos
Desktop   Downloads  Pictures  Templates
```

Step-9: ls -l command is used to show information about file.txt, including its permissions and ownership.

```
asr@fedora:~$ ls -l asr.text
-rw-r--r--. 1 asr asr 0 Sep 26 21:46 asr.text
```

Step-10: sudo chown asr2 asr.text comamand is used to change the ownership of demo1.txt to the specified user

```
asr@fedora:~$ sudo chown asr2 asr.text
[sudo] password for asr:
asr@fedora:~$ ls -l asr.text
-rwxr-xr-x. 1 asr2 asr 0 Sep 26 21:46 asr.text
```

Step-11: sudo chmod 755 asr.text command is used to set file permissions to 755 (owner can read, write, execute; group and others can read and execute).

```
asr@fedora:~$ sudo chmod 755 asr.text
asr@fedora:~$ ls -l asr.text
-rwxr-xr-x. 1 asr2 asr 0 Sep 26 21:46 asr.text
```

Step-12: sudo chmod u+rwx asr.text is used to add read and write permissions for the user on asr.txt.

```
asr@fedora:~$ sudo chmod u+rwx asr.text
asr@fedora:~$ ls -l asr.text
-rwxr-xr-x. 1 asr2 asr 0 Sep 26 21:46 asr.text
```

Step-13: sudo dnf install acf is used to install the ACL package for fine-grained permission control.

```
asr@fedora:~$ sudo dnf install acl
Fedora 40 - x86_64                              1.2 MB/s |  20 MB     00:16
Fedora 40 openh264 (From Cisco) - x86_64        460  B/s | 1.4 kB     00:03
Fedora 40 - x86_64 - Updates                    2.6 MB/s |  10 MB     00:03
Last metadata expiration check: 0:00:01 ago on Thu 26 Sep 2024 10:18:43 PM IST.
Package acl-2.3.2-1.fc40.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

Step-14: sudo setfacl -m u:asr2:rwx asr.text is used to give read permission to a specific user on demo1.txt. sudo getfacl asr.text is used to display the ACL entries for file.txt.

```
asr@fedora:~$ sudo setfacl -m u:asr2:rwx asr.text
[sudo] password for asr:
asr@fedora:~$ sudo getfacl asr.text
# file: asr.text
# owner: asr2
# group: asr
user::rwx
user:asr2:rwx
group::r-x
mask::rwx
other::r-x
```

5. Learning outcomes (What I have learned):

a) Understand how to create, modify, and delete user accounts using useradd and userdel.
b) Learn how to organize users into groups using groupadd and manage group memberships.
c) Master the use of chmod to set and modify file permissions.
d) Understand how to change group ownership of files using chgrp and how to modify file ownership using chown.

Evaluation Grid (To be created as per the SOP and Assessment guidelines by the faculty):

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Worksheet		8 Marks
2.	Viva		10 Marks
3.	Simulation		12 Marks
	Total		30 Marks

Teacher Signature